

# YUJIA LI

Lawrence Berkeley National Laboratory ◇ Grid Integration Group, ESDR Division

[yujiali@lbl.gov](mailto:yujiali@lbl.gov) ◇ +1 (510) 499-4662

[Google Scholar Profile](#)

## RESEARCH INTERESTS

---

Power system **resilience** under extreme events

**Uncertainty quantification** and optimization for renewable integration

**Transportation electrification** and EV charging infrastructure

Integrated **transmission and distribution** system planning

## EDUCATION

---

**The University of Hong Kong**

*Sept 2018 – Aug 2022*

Ph.D. in Electrical Engineering

**Advisors:** Prof. Yunhe Hou and Prof. David J. Hill

**Thesis:** Toward Sustainable and Resilient Power Infrastructure: Operation and Evaluation Strategies Under Exogenous and Endogenous Uncertainties

**Huazhong University of Science and Technology**

*Sept 2014 – June 2018*

B.E. in Electrical Engineering

**Advisor:** Prof. Shihong Miao

**Thesis:** Stochastic Security-Constrained Unit Commitment with Virtual Power Plants

## ACADEMIC APPOINTMENTS

---

**Lawrence Berkeley National Laboratory**

March 2024 – Present

*Postdoctoral Researcher*

*Berkeley, CA*

- Grid Integration Group, ESDR Division. **Advisor:** Dr. Miguel Heleno
- Develop optimization models that coordinate transmission and distribution system planning under high penetrations of renewable and distributed energy resources
- Link planning models with electricity market design to support reliable, cost-effective grid expansion

**The University of Hong Kong**

Sept 2022 – Feb 2024

*Postdoctoral Researcher*

*Hong Kong*

- Department of Electrical and Electronic Engineering. **Advisor:** Prof. Yunhe Hou
- Investigated the coupling of transportation and power systems, including planning and operation of fast charging stations, dynamic wireless charging, and high-speed rail traction loads
- Developed distributionally robust and stochastic optimization models under decision-dependent uncertainties
- Mentored two full-time Ph.D. students on topics on resilient power system operation and transportation electrification

**The University of Central Florida**

Dec 2021 – May 2022

*Visiting Scholar*

*Orlando, FL*

- Department of Electrical and Computer Engineering. **Advisor:** Dr. Wei Sun
- Research Topic: Restoration of Active Distribution Networks

## PUBLICATIONS

---

### Journal Papers

- J1 Yujia Li, Samuel Cordova, Alexandre Moreira and Miguel Heleno, “Netload Range Cost Curves for a Transmission - Aware Distribution System Planning under DER Growth Uncertainty,” *IEEE Transactions on Energy Markets, Policy and Regulation*, 2025, early access.
- J2 Yujia Li, Feng Qiu, Yixuan Chen and Yunhe Hou, “Adaptive Distributionally Robust Planning for Renewable-Powered Fast Charging Stations Under Decision-Dependent EV Diffusion Uncertainty,” *IEEE Transactions on Smart Grid*, vol. 15, no. 6, pp. 5853-5870, Nov. 2024.
- J3 Yujia Li, Shunbo Lei, Wei Sun and Yunhe Hou, “A Distributionally Robust Resilience Enhancement Strategy for Active Distribution Networks Considering Decision-dependent Contingencies,” *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1450-1465, March 2022.
- J4 Yujia Li, Wei Sun, Wenqian Yin, Shunbo Lei and Yunhe Hou, “Restoration Strategy for Active Distribution Systems Considering Endogenous Uncertainty in Cold Load Pickup,” *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 2690-2702, October 2021.
- J5 Yujia Li, Wenqian Yin, Qinfei Long and Yunhe Hou, “Review on Resilience Enhancement Strategies for Power Systems with Massive Resources Under Multi-Source Uncertainties,” *High Voltage Engineering*, vol. 09, pp. 3484-3496, 2022.
- J6 Chenxi Hu, \*Yujia Li, Yunhe Hou, “Risk-Informed Resilience Planning of Transmission Systems Against Ice Storms,” *Applied Energy*, vol. 392, pp. 125801, 2025. (\*Corresponding Author)
- J7 Ruizhang Yang, \*Yujia Li, Zhuang Xiao and Yunhe Hou, “Coordinative Operating Strategy of Grid-Interactive FTPSS-HST for Cost-Efficient Operation,” *IEEE Transactions on Smart Grid*, 2024. (\*Corresponding Author)
- J8 Ruizhang Yang, Yujia Li, Chenxi Hu, Francis Yunhe Hou, “ExDiffusion: Classifier-Guidance Diffusion Model for Extreme Load Scenario Generation with Extreme Value Theory,” *IEEE Transactions on Smart Grid*, 2025, early access.
- J9 Chenxi Hu, Shunbo Lei, Yujia Li, Yunhe Hou, “Decision-Dependent Resilience Enhancement Strategy for Distribution Systems Against Endogenous Wildfires,” *IEEE Transactions on Smart Grid*, 2025, early access.
- J10 Wenqian Yin, Yujia Li, Jiazuo Hou, Miao Miao and Yunhe Hou, “Coordinated Planning of Wind Power Generation and Energy Storage With Decision-Dependent Uncertainty Induced by Spatial Correlation,” *IEEE Systems Journal*, 2022, vol. 17, no. 2, pp. 2247-2258.
- J11 Rong-Peng Liu, Yunhe Hou, Yujia Li, Shunbo Lei, Wei Wei and Xiaozhe Wang, “Sample Robust Scheduling of Electricity-Gas Systems Under Wind Power Uncertainty,” *IEEE Transactions on Power Systems*, vol. 36, no. 6, pp. 5889-5900, Nov. 2021.
- J12 Chang Li, Jiayan Liu, Qi Liu, Yujia Li, Yong Li, “Voltage regulation in PV-rich distribution networks: an edge pipelined intelligent computing approach,” *Communications Engineering*, vol. 4, no. 1, Nov. 2025.

### Book Chapter

- B1 Yujia Li, Chenxi Hu, Yixuan Chen, Junhong Liu, Chong Wang, Rongpeng Liu, and Yunhe Hou, “Resilience of Transmission Systems,” in *Power Grid Resilience: Theory and Applications*, Springer Nature Switzerland, pp. 47-108, Jan. 2025.

## Conference Papers

- C1 **Yujia Li**, Chenxi Hu, Yunhe Hou, “The Value of Ambiguity Quantification in Distributionally Robust Economic Dispatch Models for the Wind-Penetrated Power System,” *2023 IEEE PES GT&D*, 2023.
- C2 **Yujia Li**, Wenqian Yin, Shuanglei Feng and Yunhe Hou, “Two-Stage Stochastic Unit Commitment Considering Distribution Perturbation via Contamination Technique,” *2020 IEEE Power & Energy Society General Meeting (PESGM)*, 2020, pp. 1-5.
- C3 **Yujia Li**, Shuanglei Feng and Yunhe Hou, “Assessing Robustness of Risk-Constrained Operating Strategies for Power Systems with Renewables by Contamination-Based Technique,” *2019 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*, 2019, pp. 1-5.
- C4 Yuxuan Wang, **Yujia Li**, Qinfei Long, Ruizhang Yang, Yunhe Hou, “A Wind Power Forecasting Method Using Causal Inference,” *2023 IEEE 7th Conference on Energy Internet and Energy System Integration (EI2)*, Shenyang, China, 2023, pp. 1-5.
- C5 Wenqian Yin, **Yujia Li**, Jiazuo Hou and Yunhe Hou, “The Economic Value of Improving Forecasting Accuracy in High Wind Penetrated Power Systems,” *2021 IEEE PES Innovative Smart Grid Technologies - Asia (ISGT Asia)*, Brisbane, Australia, 2021, pp. 1-5.
- C6 Li Sun, **Yujia Li**, Jiabing Hu and Yunhe Hou, “Enhanced Frequency Control for DFIG WTs to Improve Transient Power Sharing with SGs,” *2019 IEEE Sustainable Power and Energy Conference (iSPEC)*, Beijing, China, 2019, pp. 1524-1529.

## PROFESSIONAL SERVICE

---

### Leadership Roles

- **Chair**, IEEE PES Task Force on Coordinated Restoration for Resilient Multi-Energy Systems (2025 – Present)
- **Webmaster**, IEEE PES Working Group on Power System Restoration (2022 – Present)
- **Secretary**, IEEE PES Task Force on Power System Restoration with Renewable Energy Sources (2022 – 2025)

### Conference Organization

- **Panel Session Chair**, “Optimal Restoration Strategy with Security Constraints in Inverter Dominated Power Systems,” *IEEE PES General Meeting 2025*, Austin, TX
- **Panel Session Chair**, “Toward Resilient Power Systems: Advanced Methodologies for Uncertainty Management During Extreme Events,” *INFORMS Annual Meeting 2024*, Seattle, WA
- **Panel Session Chair**, “Restoration in IBR-Dominated Power Systems,” *IEEE PES General Meeting 2024*, Seattle, WA
- **Panel Session Chair**, “Enhancing Climate Resilience of Renewable-Dominated Power Grids,” *INFORMS Annual Meeting 2023*, Phoenix, AZ

### Peer Reviewer

- IEEE Transactions on Smart Grid, Power Systems, Sustainable Energy, Energy Markets Policy & Regulation
- Applied Energy, IET Smart Grid, Electric Power Systems Research, etc.

## RESEARCH PROJECTS

---

### **Co-Optimization/Simulation for Transmission and Active Distribution Networks with Microgrids**

2025 – Present

*Key Researcher*

- **Funding Source:** Department of Energy: Microgrids R&D Program
- Lead LBNL's use case with La Plata Electric Association, building an integrated T&D-microgrid planning model that delivers verifiable “no-reverse-flow” guarantees at transmission boundaries across future uncertain net-load scenarios
- Develop Netload Range Cost Curves (NRCC) to summarize high-dimensional planning decisions and uncertainty at the distribution level into tractable decision tools for transmission planners

### **Block-Level Approach for Risk Mitigation in Electricity Networks**

2023 – 2025

*Key Researcher*

- **Funding Source:** Department of Energy: Microgrids R&D Program
- Develop long-term planning optimization models for power systems, using scenario-based and robust optimization to capture spatial and temporal variability
- Design efficient computation methods to solve large-scale problem considering the coordination between power transmission and distribution systems

### **Developing Renewable-Dominated Distribution Systems with Dynamic Wireless Charging for Electric Vehicles**

2023 – 2024

*Co-PI*

- **Funding Source:** Seed Fund for PI Research – Basic Research (Hong Kong)
- Modeled dynamic wireless charging tracks under various traffic conditions and integrated them with distribution network operation
- Assessed spatio-temporal vulnerabilities of renewable-dominated distribution systems with DWCTs and proposed resilience-enhancing operating strategies

### **Bidding Strategy for Energy Storage Systems Utilizing AI-Based Forecast of Renewables and Electricity Prices**

2023 – 2024

*Technical Consultant*

- **Funding Source:** Gangke Smart Power Holding Group Co., Limited
- Developed AI-based forecasting models (LSTM-type recurrent networks) for renewable generation and electricity prices
- Integrated forecasts into game-theoretic bidding strategies for storage units in balancing and ancillary services markets
- Worked closely with industry partners to translate models into deployable tools

### **Development of Integrated Energy Management Platform for Fast EV Charging Facilities** 2023 – 2024

*Key Researcher*

- **Funding Source:** Innovation and Technology Fund (ITF), Hong Kong
- Optimize the location and size of the battery-assisted fast charger (BAFC) based on stochastic modeling of charging demands by arc cover-path cover (AC-PC) model
- Design the hardware architecture of the BAFC network and a cloud-based energy-storage-as-a-service (ESaaS) platform

## Fundamental Research on Structural Resilience of Power Systems with Incomplete Information under Severe Weather Events

2022 – 2024

Key Researcher

- **Funding Source:** National Natural Science Foundation of China (NSFC)
- Quantify the uncertain propagation patterns of extreme weather events (EWEs)
- Construct resilience enhancement strategies for power systems with incomplete grid information

## TEACHING EXPERIENCE

---

**The University of Hong Kong**

Spring 2023

Co-Instructor

- Course: Power System Analysis and Control (Undergraduate)
- Focused on fundamental optimization theory applied in power systems

**The University of Hong Kong**

Spring 2022 & Autumn 2020

Teaching Assistant

- Courses: Power System Analysis and Control; Power Transmission and Distribution
- Led weekly and bi-weekly lab sessions for undergraduate students

## SELECTED INVITED TALKS

---

- T1 29 July 2025, “Restoration Strategy for Active Distribution Systems Considering Endogenous Uncertainty in Cold Load Pickup,” *IEEE PES General Meeting*, Austin, U.S.
- T2 20 October 2024, “Toward Resilient Power Systems: Advanced Methodologies for Uncertainty Management During Extreme Events,” *INFORMS Annual Meeting*, Seattle, U.S.
- T3 23 July 2024, “Restoration in IBR-Dominated Power Systems,” *IEEE PES General Meeting*, Seattle, U.S.
- T4 14 October 2023, “Enhancing Climate Resilience of Renewable-Dominated Power Grids,” *INFORMS Annual Meeting*, Phoenix, U.S.
- T5 18 July 2023, “Enhancing Resilience of Grid-Interactive Efficient Buildings Using Reinforcement Learning,” *IEEE PES General Meeting*, Orlando, U.S.
- T6 18 October 2022, “A Distributionally Robust Resilience Enhancement Strategy for Distribution Grids Considering Decision-Dependent Contingencies,” *INFORMS Annual Meeting*, Indianapolis, U.S.
- T7 17 July 2022, “Restoration Strategy for Active Distribution Systems Considering Endogenous Uncertainty in Cold Load Pickup,” *IEEE PES General Meeting*, online
- T8 24 October 2021, “Stochastic Restoration Strategy Considering Decision-dependent Cold Load Pickup,” *INFORMS Annual Meeting*, online
- T9 03 August 2020, “Two-Stage Stochastic Unit Commitment Considering Distribution Perturbation via Contamination Technique,” *IEEE PES General Meeting*, online
- T10 30 September 2019, “Assessing Robustness of Risk-Constrained Operating Strategies for Power Systems with Renewables by Contamination-Based Technique,” *IEEE Innovative Smart Grid Technologies - Europe*, Bucharest, Romania
- T11 22 May 2019, “A MILP Production Costing Method Realizing Environmental Benefits of Wind Power,” *IEEE Innovative Smart Grid Technologies - Asia*, Chengdu, China